

CPP Program

```
namespace ns1
{
    int num1=30;
}

int num1=20; // global
// int num1=30; // name collisions

int main()
{
    int num1=10; // local
    // int num1=20; // not ok ⇒ name collisions
    printf("\n num1=%d", num1); // 10
}
```

logical division/grouping
namespace

global namespace

Demol.cpp

```
#include<stdio.h>

namespace ns1
{
    int num1=30; // elemnt of namespace
    int num2=40; // elemnt of namespace

    void hi() // elemnt of namespace
    {
        printf("\n Hi Students ... !");
    }

    // element of namespace ns1
    namespace nesNs // nested namespace ⇒ namespace within namespace
    {
        int num1=50; // element of nested namespace
    }
}

int num1=20; // global
// int num1=30; // name collisions
int main()
{
    int num1=10; // local
    // int num1=20; // not ok ⇒ name collisions
    printf("\n num1=%d", num1); // 10 // priority will be given to local var
    // Q. how to access an element of namespace
    // ans. namespace::element
    printf("\n ns1 num1=%d", ns1::num1); // 30 // access an element of namespace
    printf("\n ns1 num2=%d", ns1::num2); // 40 // access an element of namespace
    ns1::hi(); // calling a function which is an element of namespace called ns1

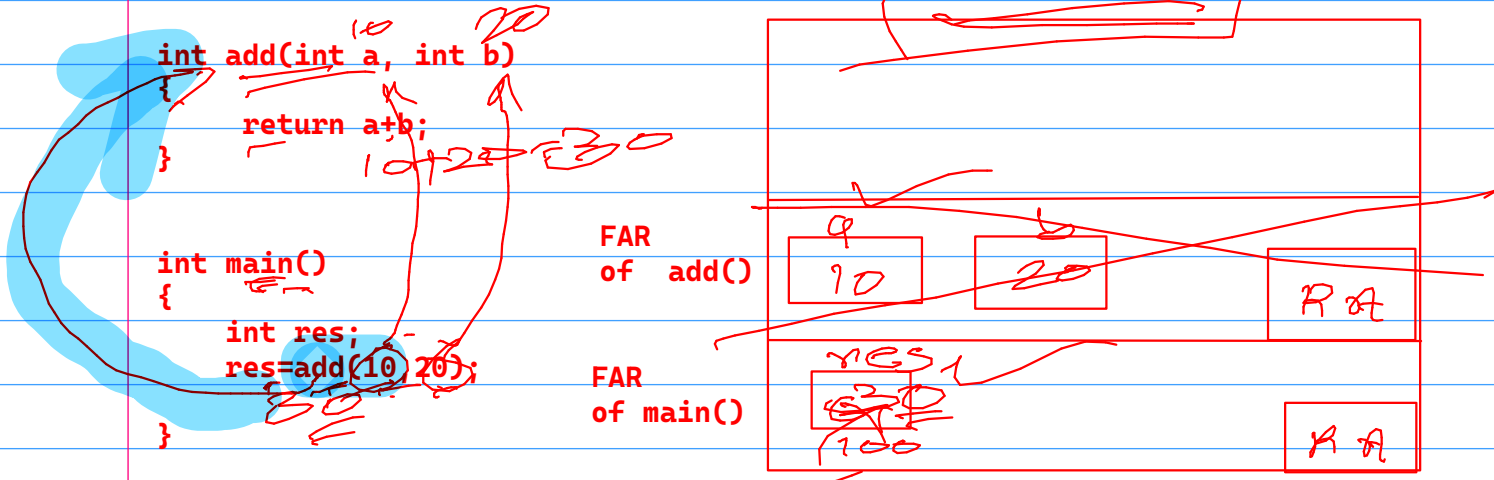
    // How to access an element fo nested namespace?
    printf("\n nes namespace ele num1=%d", ns1::nesNs::num1);
}
```

ns1
namespace

nested
namespace

global
namespace

Function



inline function

```
inline int add(int a, int b)
{
    return a+b;
}
```

```
int main()
{
    int res;
    res=add(10,20); // when we call the inline function
}
```

default argument

```
void sum(int n1, int n2, int n3=0, int n4=0)
{
    cout<<"sum="<<n1+n2+n3+n4<<endl;
}
// 10+ 20+30+0=>60 //10+20+0+0 =>30
int main()
{
    // sum(10,20,30,40,50); //too many arguments in function call
    sum(10,20,30); //too few arguments in function call
    sum(10,20);
}
```

Handwritten annotations in the diagram:

- Arguments 10 and 20 are passed to `n1` and `n2` respectively.
- The default values 0 and 0 are circled and labeled "default value".
- The calculation `10+ 20+30+0=>60` is shown for the `sum(10,20,30)` call.
- The calculation `10+20+0+0 =>30` is shown for the `sum(10,20)` call.

Batch PH38

C1 :: Sohan

Sohan

C1

Sohan

C2

C2 :: Sohan

Sohan

C3

C4 :: Sohan

Sohan

C4

namespace

ns1::num1;

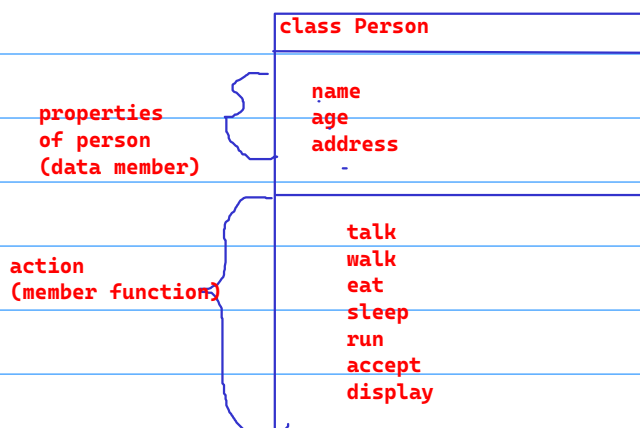
```
#include<stdio.h>

namespace ns1
{
    int num1=30;
}

int num1=20; //global var
//name collision
int main()
{
    int num1=10;
    printf("num1=%d", num1); //10
}
```

logical division or grouping
namespace

logical division or grouping
global namespace



stack area

Person p1; //p1 is an obj of class Person

p1.accept();
p1.display();

p1 obj of person class

